

- 5 (a) The body length of the medicinal leech decreases when touched.



(Source: © Mit Kapevski/Shutterstock)

The body length of the leech decreases less each time the leech is touched.

- (i) What is the name of this effect?

(1)

- ☐ **A** absorption
- ☐ **B** adaptation
- ☐ **C** habituation
- ☐ **D** sensitivity

- (ii) Why is it an advantage for the leech to respond in this way?

(1)

- ☐ **A** it filters out what is important and what is not important to react to
- ☐ **B** it enables the animal to react more quickly to the stimulus
- ☐ **C** it keeps the habitat of the animal protected
- ☐ **D** it helps the animal find food more easily

(b) The photograph shows a fiddler crab.



(Source: © Jay Gao/Shutterstock)

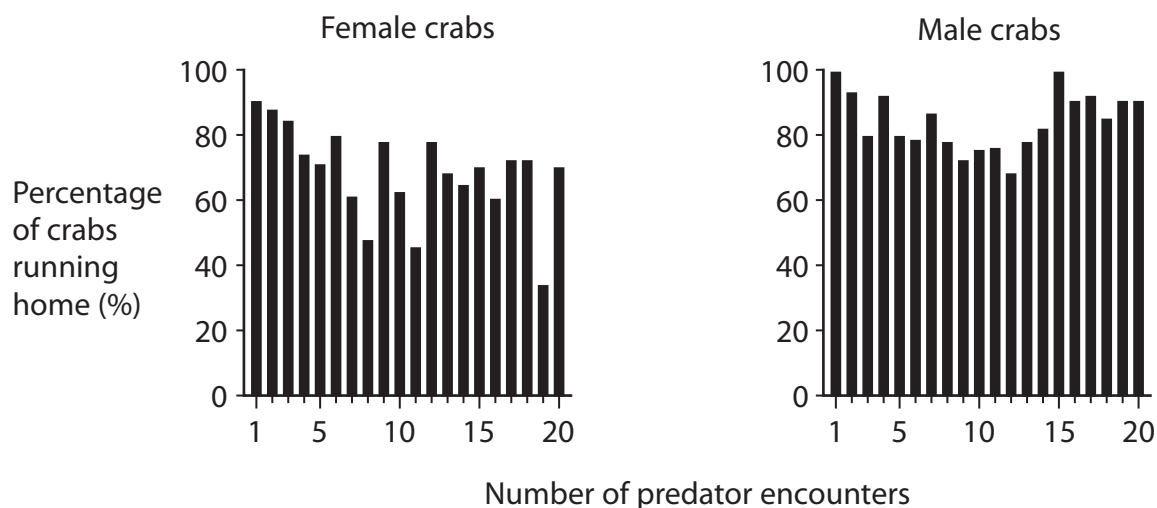
In an experiment, the fiddler crab was used to investigate responses to a predator. The responses by females and males were compared.

A predator approached a group of female crabs from one direction.

The percentage of crabs running home was recorded. This was repeated 20 times.

This experiment was repeated with a group of male crabs.

The graphs show the percentage of crabs running home for both groups of crabs against the total number of times each group of crabs encountered a predator.



(i) Comment on the results of this experiment.

(3)

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(ii) Which statistical test would be used to analyse the effect of the increase in the number of predator encounters on the percentage of crabs running home?

(1)

- ☐ **A** correlation coefficient
- ☐ **B** Hardy-Weinberg
- ☐ **C** index of diversity
- ☐ **D** Student's t-test

(Total for Question 5 = 6 marks)